

VIGNESHVARAN M

Electrical & Electronics Engineering Graduate (B.E. EEE, 2026)
+91 63833 84049 | varanv160@gmail.com | Dindigul, Tamil Nadu – 624402
Date of Birth: 25-11-2004 | Nationality: Indian | Languages: Tamil, English
Willing to relocate | Available to join immediately

EDUCATION

B.E., Electrical and Electronics Engineering	2022 – 2026
NPR College of Engineering & Technology, Natham CGPA: 7.91 (First Class)	
Higher Secondary (HSC)	2022
Sarva Seva Higher Secondary School, Ayyanarpuram 65.16% (First Class)	
SSLC	2020
Sarva Seva High School, Andipuram 82.6% (First Class)	

CAREER OBJECTIVE

Electrical engineering graduate with industry training in EV and battery systems, power transmission and PLC automation, a granted patent, and national-level innovation recognition. Seeking an entry-level role to apply skills in electrical system design, testing and renewable energy, and to grow within a global engineering organisation.

INTERNSHIPS & TRAINING

- **Successfully completed a 5-month engineering internship in Transenergy Private Limited, Chennai**, gaining hands-on exposure to electrical systems, equipment operation, maintenance practices, troubleshooting, and industrial safety procedures.
- **In-Plant Training at TEZTA Electric Mobility & Energy Pvt. Ltd.** – Gained hands-on experience with electric vehicles and lithium-ion battery technologies, including design, operation, and maintenance.
- **In-Plant Training at TANTRANSCO Limited (TNEB)** – Gained practical exposure to power transmission systems, substation operations, and electrical network management.
- **Industrial Automation Training – PLC Programming at TVS Training and Services Ltd. (5 days)**, gained hands-on experience in programmable logic controllers and automation systems.

PROJECTS

- **TIDCO Project: “Development of a Smart Modular and Interoperable EV Energy Ecosystem with AI-Driven Battery Intelligence for Electric Two-Wheelers”** – Participated as part of the project team, presented the project before a jury, and was selected as the Best Project.
- **Final Year Project:** Designed and developed a Smart In-Pipe Energy Harvesting System focused on converting energy from fluid flow inside pipelines into usable electrical energy. The system delivered a 12 V output, demonstrated in an agricultural pump house with a 15 V battery for storage.
- **Project: Solar Powered IoT Agri-Vehicle for Precision Farming** – Developed an innovative solar-powered agriculture vehicle integrated with IoT technology to enhance precision farming, improve efficiency, and promote sustainable agricultural practices.
- **Showcased Project on Science Day at NPR College of Engineering and Technology** – “Solar Powered IoT Agri-Vehicle for Precision Farming”, demonstrating innovative IoT-based solutions for sustainable and precision agriculture.
- **Participated in Project Competition at Velammal College of Engineering and Technology** – “Inside Drainage Monitoring and Cleaning”, developing innovative solutions for smart drainage maintenance and urban sanitation.
- **Mini Project at NPR College of Engineering and Technology** – “Live Current Detector”, developed a device to detect live electrical currents safely, enhancing understanding of electrical safety and practical circuit design.

ACHIEVEMENTS & AWARDS

- **Best Project:** Selected as the Best Project after presenting the TIDCO EV Energy Ecosystem project before a jury.
- **Patent:** Granted for an IoT-Based Solar Powered Agri Vehicle (R435223-001), for the innovative design and application of IoT-enabled solar-powered technology in agricultural robotics.
- **Selected among the Top 10 finalists at the National-Level IPR Innovation Challenge** for the Smart In-Pipe Energy Harvesting System, demonstrating innovation in energy harvesting and engineering system development.
- **Won First Prize in Paper Presentation at Sethu Institute of Technology, Karaipatti**, on “IoT-Based Solar Agri Vehicle”, showcasing innovation in renewable energy and smart farming applications.

TECHNICAL SKILLS

Domain: Electric vehicles, electrical machines, power systems, renewable energy systems, PLC automation, PCB design

Software: MATLAB, PowerWorld Simulator, MS Word, MS PowerPoint

Programming: Java (OOP, collections, multithreading, JDBC), HTML, CSS, JavaScript

Soft Skills: Team leadership, problem solving, time management, quick learner

ACTIVITIES

- **Participated in Paper Presentation at Kongu Engineering College, Erode** on “Multi Drive System for EV”, demonstrating research and innovative approaches in electric vehicle technology.
- **Shortlisted in Smart India Hackathon 2024 – Paper Presentation** for “Gravity Driven Solar Tracking”, recognized for innovative design to improve solar energy efficiency and selected for the next round.
- **Participated in Techgium 8th Edition – Paper Presentation** on “Automatic Compensation of Torque Gaps in Electric Vehicles on Rough Terrain”, demonstrating innovative solutions for EV performance and gaining exposure to technical research and problem-solving.
- **Participated in Paper Presentation at L&T** on “Inside Pipe Hydro Power Generation”, showcasing innovative approaches to renewable energy and gaining exposure to practical engineering applications and technical research.
- **Submitted Ideas in MSME Idea Hackathon 4.0 and 5.0** on “IoT-Based Solar Agri Vehicle” and “Inside Pipe Hydro Power Generation”, presenting innovative approaches to sustainable agriculture, renewable energy and power generation.
- Submitted abstract and was shortlisted for the next level of presentation; participated in Design & Innovation Clinic 2025 at Central Manufacturing Technology Institute, Bengaluru, under the Indian Government, presenting “IoT-Based Solar Agri Vehicle for Precision Farming” and demonstrating innovative solutions in sustainable agriculture and precision engineering.
- **Hands-on Training in PCB Design:** Attended practical training on PCB design to gain skills in circuit layout, component placement, and routing using modern design software for electronic hardware development.
- **Hands-on Training: Bicycle to Electric Vehicle Conversion –** Attended practical training on converting a conventional bicycle into an electric vehicle, gaining knowledge in motor integration, battery management, controller setup, and wiring systems.

WORKSHOPS

- **Additive Manufacturing Training – PSG Institute of Technology**, gained hands-on experience in advanced 3D printing and manufacturing.
- **e-Yantra Robocon Rebooted Applied Research**, participated in applied robotics research projects, enhancing skills in automation, programming, and problem-solving.
- **Smartphone Debugging and Troubleshooting – NPR College of Engineering and Technology**, gained hands-on experience in identifying and resolving hardware and software issues in mobile devices.

CONFERENCES

- Participated in a technical engineering conference at Velammal Institute of Technology, Chennai, gaining exposure to emerging technologies, industrial practices, and innovative engineering solutions.
- Participated in the International Conference on Smart Innovation in Engineering & Technology at NPR College of Engineering and Technology.

INDUSTRIAL VISITS

- **Industrial Visit – Windcare India Pvt. Ltd.**, observed operations and technologies in the renewable energy sector.
- **Industrial Visit – ARKAY Power Plant, Kulathur, Ramanathapuram**, gained exposure to power plant operations, energy generation processes, and industrial safety practices.
- **Industrial Visit – ZF Rane Automotive India Pvt. Ltd.**, learned about the manufacturing of steering and suspension components, observing design, machining, assembly, quality control, automation and lean manufacturing practices.
- **Industrial Visit – Aqua Sub Engineering**, studied Texmo borewell submersible and single-phase domestic pumps, and gained insights into the manufacturing, testing, and quality control of single-phase jet pumps.

PLACEMENT TRAINING

- Aspiring Full Stack Developer, currently undergoing a one-month placement training program with hands-on experience in building web applications using HTML, CSS, and JavaScript. Skilled in designing responsive front-end interfaces; completed projects.
- Completed 15-day placement training in Java, covering core concepts such as OOP, exception handling, collections, multithreading, and JDBC, with hands-on practice through mini-projects and console-based application

WORKING EXPERIENCE

- Electrical Maintenance – Aqua sub Engineering,Coimbatore perform electrical maintenance,troubleshooting,and fault rectification of industrial machines and control panels.
- Handle motors, contactors, relays,MCB and electrical wiring and protection system